

DIFFERENTIAL EQUATIONS

Complete Lecture Notes — Lectures 1 to 8

Lecture 1 — Introduction

Branches of Calculus

Differential Calculus	Integral Calculus
Differential: independent change in variable. Derivative: change w.r.t. something (dependent change). <i>The process of finding derivatives is called differentiation.</i>	- Find the area under a curve. - Find the original function from its derivative.

Physical Phenomena and Mathematical Modelling

Physical phenomena (especially time-dependent ones) are described via mathematical models, which lead to the **Theory of Differential Equations**. The pipeline is:

Physical Phenomenon $\rightarrow$ Mathematical Model $\rightarrow$ Theory of DEs $\rightarrow$ Simplify $\rightarrow$ Solution $\rightarrow$ Physics

DEs provide *short-term information* about a system. The key steps are: **IDENTIFY** $\rightarrow$ **SIMPLIFY**.

Types of Differential Equations

ODE (Ordinary DE)	PDE (Partial DE)
Contains ordinary derivatives (d/dx). Involves one independent variable. Models 1-D phenomena.	Contains partial derivatives ($\partial/\partial x$). Involves more than one independent variable. Models n-D phenomena.

Lecture 2 — Definitions

Differential Equation

Definition: Differential Equation

An equation involving a **dependent variable** and its **derivative(s)** with respect to one or more independent variables.

Ordinary Differential Equation (ODE)

Definition: ODE

A differential equation that contains **ordinary derivatives** of the dependent variable with respect to **exactly one** independent variable.

Examples of ODEs:

$$\frac{dy}{dx} + y \cos x = \sin x \quad (1 \text{ — 1st order})$$

$$\frac{d^2y}{dx^2} + xy \left(\frac{dy}{dx} \right)^2 = 0 \quad (2 \text{ — 2nd order})$$

$$\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2} = \frac{d^2y}{dx^2} \quad (3 \text{ — 2nd order})$$

Partial Differential Equation (PDE)

Definition: PDE

A differential equation that contains **partial derivatives** of a dependent variable with respect to **more than one** independent variable.

Examples of PDEs:

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu \quad (4 \text{ — 2-D PDE})$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0 \quad (5 \text{ — 3-D Laplace PDE})$$

Order and Degree

Definition: Order

The **order** of a DE is the order of the **highest derivative** that appears in the equation.

Definition: Degree

The **degree** of a DE is the **greatest exponent** of the highest-order derivative in the equation, after clearing all radicals and fractions.

Note: The dependent variable and its derivatives must be expressed free from radicals and fractions before determining the degree.

Order and Degree Table:

Equation	Order	Degree
$\frac{dy}{dx} + y \cos x = \sin x$	1	1
$\frac{d^2y}{dx^2} + xy \left(\frac{dy}{dx} \right)^2 = 0$	2	1
$\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2} = \frac{d^2y}{dx^2}$	2	2 (after squaring)
$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$	1	1
$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$	2	1

Linear vs. Non-Linear DEs

Definition: Linear DE

A DE is **linear** if **all** of the following hold:

1. The degree of the DE is 1.
2. There is **no product** of the dependent variable and/or its derivatives.
3. **No transcendental** function (trigonometric, exponential, logarithmic) of the dependent variable appears.

If *any* condition is violated, the DE is **non-linear**.

Solutions of a DE

Definition: Solution (Integral) of a DE

A relation between variables (dependent and independent) that contains **no derivatives**, such that when derivatives are computed from it, the relation **identically satisfies** the given DE.

- **General Solution:** Contains as many arbitrary constants as the *order* of the DE.
- **Particular Solution:** Obtained from the general solution by assigning specific values to the arbitrary constants (via initial/boundary conditions).

$$\boxed{\text{Number of arbitrary constants} = \text{Order of the DE}}$$

Lecture 3 — ICs, BCs, Direction Fields and ODE Types

Initial and Boundary Conditions

Initial Conditions (ICs)	Boundary Conditions (BCs)
All conditions specified at one value of the independent variable. <i>Example:</i> $y(0) = 1$, $y'(0) = 2$, $y''(0) = 4$	Conditions specified at more than one value of the independent variable. <i>Example:</i> $y(0) = 1$, $y(1) = 2$

- **Initial Value Problem (IVP):** DE + Initial Conditions
- **Boundary Value Problem (BVP):** DE + Boundary Conditions

Eliminating Arbitrary Constants (Forming DEs)

To form the DE corresponding to a general solution with n arbitrary constants: differentiate n times, then eliminate the constants.

Example: Find the DE for $y = A \sin x + B \cos x$ (2 constants $\Rightarrow$ 2nd-order DE)

$$\begin{aligned} y &= A \sin x + B \cos x \\ y' &= A \cos x - B \sin x \\ y'' &= -A \sin x - B \cos x = -y \end{aligned} \tag{I}$$

$$\boxed{\frac{d^2 y}{dx^2} + y = 0}$$

Direction Fields

- Direction fields give a **graphical interpretation** of a DE *without solving it*.
- At each point (x, y) a short line segment is drawn with slope dy/dx from the DE.
- Solution curves follow these slope segments.

Types of First-Order ODEs (Overview)

1. Separable ODEs
2. Reducible to Separable ODEs (Homogeneous substitution)
3. Exact ODEs
4. Reducible to Exact ODEs (Integrating Factors)
5. Linear ODEs

Lecture 4 — Separable ODEs

Definition

Definition: Separable ODE

A DE of the form

$$\frac{dy}{dx} = \frac{F(x)}{G(y)} \iff G(y) dy = F(x) dx$$

in which all x -terms can be grouped on one side and all y -terms on the other.

Method of Solution

Steps to Solve a Separable ODE

1. Write in the form $\frac{dy}{dx} = \frac{F(x)}{G(y)}$.
2. Separate variables: $G(y) dy = F(x) dx$.
3. Integrate both sides: $\int G(y) dy = \int F(x) dx + C$.
4. Simplify and solve for y (general solution).
5. If an IVP is given, substitute the initial condition to find C .

Important: The arbitrary constant C must be introduced *immediately* during integration. If taken as $y = e^{\lambda x + C}$, it must be rewritten as $y = C_1 e^{\lambda x}$ where $C_1 = e^C$, to ensure the solution is valid for all non-zero C .

Worked Examples

Example 1: $\frac{dy}{dx} = 2xy$

$$\begin{aligned}\frac{1}{y} dy &= 2x dx \\ \int \frac{1}{y} dy &= \int 2x dx \\ \ln |y| &= x^2 + C \\ y &= e^{x^2 + C} = e^C \cdot e^{x^2}\end{aligned}$$

$$\boxed{y = C_1 e^{x^2}, \quad C_1 = e^C}$$

Example 2: $y^3 \frac{dy}{dx} + x^3 = 0$

$$\begin{aligned}y^3 dy &= -x^3 dx \\ \int y^3 dy &= \int -x^3 dx \\ \frac{y^4}{4} &= -\frac{x^4}{4} + C\end{aligned}$$

$$\boxed{x^4 + y^4 = C_1}$$

Example 3 (Homogeneous Substitution): $x \frac{dy}{dx} = y + 2x^2 \sin^2\left(\frac{y}{x}\right)$, IVP $y(1) = 0$

Step 1. Divide by x :

$$\frac{dy}{dx} = \frac{y}{x} + 2x \sin^2\left(\frac{y}{x}\right)$$

Step 2. Let $u = y/x$, so $y = ux$ and $\frac{dy}{dx} = u + x \frac{du}{dx}$:

$$u + x \frac{du}{dx} = u + 2x \sin^2 u \implies x \frac{du}{dx} = 2x \sin^2 u \implies \frac{du}{dx} = 2 \sin^2 u$$

Step 3. Separate and integrate:

$$\frac{du}{\sin^2 u} = 2x dx \implies \int \csc^2 u du = \int 2x dx \implies -\cot u = x^2 + C$$

$$\cot u = -x^2 + C_1 \implies u = \cot^{-1}(C_1 - x^2) \implies \frac{y}{x} = \cot^{-1}(C_1 - x^2)$$

$$y = x \cot^{-1}(C_1 - x^2)$$

Apply IC $y(1) = 0$: $0 = 1 \cdot \cot^{-1}(C_1 - 1) \Rightarrow \cot^{-1}(C_1 - 1) = 0 \Rightarrow C_1 - 1 = \cot(0) \Rightarrow C_1 = 1$.

$$y = x \cot^{-1}(1 - x^2)$$

Example 4 (IVP): $x \frac{dy}{dx} + y = 0, \quad y(4) = 6$

$$x \frac{dy}{dx} = -y$$

$$\frac{1}{y} dy = -\frac{1}{x} dx$$

$$\ln |y| = -\ln |x| + C$$

$$xy = C_1 \quad (C_1 = e^C)$$

Apply IC: $4 \cdot 6 = C_1 \Rightarrow C_1 = 24$.

$$y = \frac{24}{x}$$

Lecture 5 — Exact ODEs

Total Differential

For a function $u(x, y)$, the total differential is:

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

If $u(x, y) = c$ (constant) then $du = 0$, giving:

$$\frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = 0 \quad \Longleftrightarrow \quad M(x, y) dx + N(x, y) dy = 0$$

where $M = \partial u / \partial x$ and $N = \partial u / \partial y$.

Second-Order Partial Derivatives

For $u = u(x, y)$ we have four 2nd-order partial derivatives:

$$u_{xx} = \frac{\partial}{\partial x} \frac{\partial u}{\partial x}, \quad u_{xy} = \frac{\partial}{\partial y} \frac{\partial u}{\partial x}, \quad u_{yx} = \frac{\partial}{\partial x} \frac{\partial u}{\partial y}, \quad u_{yy} = \frac{\partial}{\partial y} \frac{\partial u}{\partial y}$$

Clairaut's Theorem (Key Result)

If $u(x, y)$ and its partial derivatives are **continuous**, then:

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x} \quad \text{i.e.,} \quad u_{xy} = u_{yx}$$

Exact ODE — Test and Solution

Definition: Exact ODE

The DE $M(x, y) dx + N(x, y) dy = 0$ is **exact** if and only if:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Solution Formula for Exact ODE

$$\int M(x, y) dx + \int [\text{terms of } N \text{ free from } x] dy = C$$

Worked Examples

Example 1: $(y + 2xy^2) dx + (x + 2x^2y) dy = 0$

$$\begin{aligned} M &= y + 2xy^2, & N &= x + 2x^2y \\ \frac{\partial M}{\partial y} &= 1 + 4xy = \frac{\partial N}{\partial x} && \checkmark \text{ EXACT} \\ \int (y + 2xy^2) dx &= xy + x^2y^2 \end{aligned}$$

No terms in N are free from x , so:

$$\boxed{xy + x^2y^2 = C}$$

Example 2: $2xy dx + x^2 dy = 0$

$$\begin{aligned} \frac{\partial M}{\partial y} &= 2x = \frac{\partial N}{\partial x} && \checkmark \text{ EXACT} \\ \int 2xy dx &= x^2y \Rightarrow \boxed{x^2y = C} \end{aligned}$$

Lecture 6 — Non-Exact ODEs & Integrating Factors

Integrating Factor (IF)

If $\partial M/\partial y \neq \partial N/\partial x$, the DE is **not exact**. We multiply by an **Integrating Factor** (IF) — also called the *magic factor* — to make it exact.

Rule 1: IF depends on x only

$$\text{If } \frac{M_y - N_x}{N} = f(x) \quad (\text{function of } x \text{ alone}) \quad \text{then} \quad \text{IF} = e^{\int f(x) dx}$$

Rule 2: IF depends on y only

$$\text{If } \frac{N_x - M_y}{M} = f(y) \quad (\text{function of } y \text{ alone}) \quad \text{then} \quad \text{IF} = e^{\int f(y) dy}$$

Worked Examples

Example 1: $-y dx + x dy = 0$

$M = -y$, $N = x$. $\partial M/\partial y = -1 \neq 1 = \partial N/\partial x \Rightarrow$ NOT exact.

Rule 1:

$$\frac{M_y - N_x}{N} = \frac{-1 - 1}{x} = \frac{-2}{x} = f(x) \Rightarrow \text{IF} = e^{\int -2/x dx} = e^{-2 \ln x} = x^{-2} = \frac{1}{x^2}$$

Multiply through by $1/x^2$:

$$-\frac{y}{x^2} dx + \frac{1}{x} dy = 0$$

Now $\partial M_{\text{new}}/\partial y = -1/x^2 = \partial N_{\text{new}}/\partial x \Rightarrow$ EXACT.

$$\int \left(-\frac{y}{x^2}\right) dx = \frac{y}{x} \Rightarrow \boxed{\frac{y}{x} = C \text{ i.e., } y = Cx}$$

Example 2: $2x \tan y dx + \sec^2 y dy = 0$

$M = 2x \tan y$, $N = \sec^2 y$. $M_y = 2x \sec^2 y$, $N_x = 0 \Rightarrow$ NOT exact.

Rule 1:

$$\frac{M_y - N_x}{N} = \frac{2x \sec^2 y}{\sec^2 y} = 2x = f(x) \Rightarrow \text{IF} = e^{\int 2x dx} = e^{x^2}$$

Multiply through by e^{x^2} and verify exactness. Solution:

$$\int 2xe^{x^2} \tan y dx = e^{x^2} \tan y \Rightarrow \boxed{e^{x^2} \tan y = C}$$

Example 3 (Q#15): $(ax + by) dx + (kx + ly) dy = 0$

For exactness: $\partial M/\partial y = b$ and $\partial N/\partial x = k$.

$$\text{Exact} \iff b = k$$

When $b = k$, the solution is:

$$\int (ax + by) dx + \int ly dy = C \Rightarrow \boxed{\frac{ax^2}{2} + bxy + \frac{ly^2}{2} = C}$$

Lecture 7 — Linear ODEs & Bernoulli Equations

Linear ODE — Standard Form

Definition: Linear ODE

$$\frac{dy}{dx} + P(x)y = Q(x)$$

- If $Q(x) = 0$: **Homogeneous** linear ODE.
- If $Q(x) \neq 0$: **Non-Homogeneous** linear ODE.

Method of Solution

Integrating Factor Method for Linear ODE

1. Write in standard form: $\frac{dy}{dx} + P(x)y = Q(x)$.
2. Find IF = $e^{\int P(x) dx}$.
3. Multiply both sides by IF. LHS becomes an exact differential:

$$\frac{d}{dx} [\text{IF} \cdot y] = \text{IF} \cdot Q(x)$$

4. Integrate both sides:

$$\text{IF} \cdot y = \int [\text{IF} \cdot Q(x)] dx + C$$

5. Solve for y .

Worked Examples

Example 1: $\frac{dy}{dx} + \frac{3}{x}y = 6x^2$

$P(x) = 3/x \Rightarrow \text{IF} = e^{\int 3/x dx} = e^{3 \ln x} = x^3.$

Multiply by x^3 :

$$x^3 \frac{dy}{dx} + 3x^2 y = 6x^5 \quad \Rightarrow \quad \frac{d}{dx}(x^3 y) = 6x^5$$

Integrate:

$$x^3 y = x^6 + C \quad \Rightarrow \quad \boxed{y = x^3 + Cx^{-3}}$$

Example 2: $x \frac{dy}{dx} = 2y + x^3 e^x$

Step 1. Rearrange to standard form. Divide both sides by x :

$$\frac{dy}{dx} = \frac{2y}{x} + x^2 e^x \quad \Rightarrow \quad \frac{dy}{dx} - \frac{2}{x}y = x^2 e^x$$

Step 2. Find IF. $P(x) = -2/x$:

$$\text{IF} = e^{\int -2/x dx} = e^{-2 \ln x} = x^{-2} = \frac{1}{x^2}$$

Step 3. Multiply through by $1/x^2$:

$$\frac{1}{x^2} \frac{dy}{dx} - \frac{2}{x^3} y = e^x \quad \Rightarrow \quad \frac{d}{dx} \left(\frac{y}{x^2} \right) = e^x$$

Step 4. Integrate both sides:

$$\frac{y}{x^2} = \int e^x dx = e^x + C$$

$$\boxed{y = x^2 e^x + Cx^2}$$

Bernoulli Equation

Definition: Bernoulli Equation

$$\frac{dy}{dx} + P(x)y = Q(x)y^n, \quad n \neq 0, 1$$

This is **non-linear** due to the y^n term, but can be **reduced to a linear ODE**.

Reduction of Bernoulli Equation to Linear ODE

Step I. Divide both sides by y^n :

$$y^{-n} \frac{dy}{dx} + P(x)y^{1-n} = Q(x)$$

Step II. Substitute $v = y^{1-n}$, so $\frac{dv}{dx} = (1-n)y^{-n} \frac{dy}{dx}$, giving $y^{-n} \frac{dy}{dx} = \frac{1}{1-n} \frac{dv}{dx}$:

$$\frac{dv}{dx} + (1-n)P(x)v = (1-n)Q(x) \quad [\text{Linear ODE in } v]$$

Step III. Solve using the IF method, then back-substitute $v = y^{1-n}$.

Example: $\frac{dy}{dx} + y = xy^3 \quad [n = 3]$

Step I. Divide by y^3 :

$$y^{-3} \frac{dy}{dx} + y^{-2} = x$$

Step II. Let $v = y^{-2}$, so $\frac{dv}{dx} = -2y^{-3} \frac{dy}{dx}$:

$$-\frac{1}{2} \frac{dv}{dx} + v = x \implies \frac{dv}{dx} - 2v = -2x \quad [\text{Linear in } v]$$

$$P = -2 \Rightarrow \text{IF} = e^{-2x}.$$

$$\frac{d}{dx}(v e^{-2x}) = -2x e^{-2x}$$

Integrate by parts: let $u = x$, $dw = e^{-2x} dx \Rightarrow w = -\frac{1}{2}e^{-2x}$:

$$\int -2x e^{-2x} dx = -2 \left[-\frac{x e^{-2x}}{2} + \int \frac{e^{-2x}}{2} dx \right] = -2 \left[-\frac{x e^{-2x}}{2} - \frac{e^{-2x}}{4} \right] = x e^{-2x} + \frac{e^{-2x}}{2}$$

So:

$$v e^{-2x} = x e^{-2x} + \frac{e^{-2x}}{2} + C$$

Divide through by e^{-2x} :

$$v = x + \frac{1}{2} + C e^{2x}$$

Back-substitute $v = y^{-2}$:

$$y^{-2} = x + \frac{1}{2} + C e^{2x}$$

Lecture 8 — Orthogonal Trajectories

Definition

Definition: Orthogonal Trajectories

Two families of curves are **orthogonal trajectories** of each other if every curve from one family intersects every curve from the other at **right angles** (90°).

Classic examples:

- $x^2 + y^2 = c$ (concentric circles) and $y = kx$ (lines through origin).
- $x^2 - y^2 = c$ (hyperbolas) and $xy = k$ (rectangular hyperbolas).

Test for Perpendicularity

Condition: Perpendicular Slopes

Two curves are perpendicular at their intersection when:

$$m_1 \cdot m_2 = -1 \quad \implies \quad m_2 = -\frac{1}{m_1}$$

where m_1, m_2 are the slopes of the tangents at that point.

Method: Finding Orthogonal Trajectories

Steps to Find OTs of the family $f(x, y, c) = 0$

1. Differentiate $f(x, y, c) = 0$ and eliminate c to get $\frac{dy}{dx} = F(x, y)$.
2. Replace $\frac{dy}{dx}$ by $-\frac{1}{F(x, y)}$ (perpendicular slope condition).
3. Solve the new DE to obtain $g(x, y, k) = 0$ — the family of orthogonal trajectories.

Worked Examples

Example 1: Find OTs of $x^2 - y^2 = c$

Step 1. Differentiate: $2x - 2y\frac{dy}{dx} = 0 \implies \frac{dy}{dx} = \frac{x}{y}$.

Step 2. Replace slope: $\frac{dy}{dx} = -\frac{y}{x}$.

Step 3. Solve:

$$\frac{1}{y} dy = -\frac{1}{x} dx \quad \implies \quad \ln|y| = -\ln|x| + C \quad \implies \quad y = \frac{k}{x} \quad \implies \quad \boxed{xy = k}$$

The rectangular hyperbolas $xy = k$ are the orthogonal trajectories of $x^2 - y^2 = c$.

Example 2: Find OTs of $y = x^2 + c$ (parabolas)

Step 1. $\frac{dy}{dx} = 2x$.

Step 2. Replace: $\frac{dy}{dx} = -\frac{1}{2x}$.

Step 3. Solve:

$$\int dy = \int -\frac{1}{2x} dx \Rightarrow \boxed{y = -\frac{1}{2} \ln |x| + K}$$

Summary Table — ODE Types and Solution Methods

Type	Form / Test	Solution Method
Separable ODE	$dy/dx = F(x)/G(y)$	Separate vars, integrate both sides
Exact ODE	$Mdx + Ndy = 0; M_y = N_x$	$\int M dx + \int N_{\text{free}} dy = C$
Non-Exact (IF in x)	$(M_y - N_x)/N = f(x)$	IF $= e^{\int f(x) dx}$, then exact method
Non-Exact (IF in y)	$(N_x - M_y)/M = f(y)$	IF $= e^{\int f(y) dy}$, then exact method
Linear ODE	$y' + P(x)y = Q(x)$	IF $= e^{\int P dx}$; integrate $d(\text{IF} \cdot y)/dx$
Bernoulli Eq.	$y' + Py = Qy^n, n \neq 0, 1$	Sub $v = y^{1-n} \rightarrow$ linear in v
Homogeneous ODE	$dy/dx = f(y/x)$	Sub $u = y/x \rightarrow$ separable in u, x
Orthogonal Trajectories	Family $f(x, y, c) = 0$	Find slope, replace $dy/dx \rightarrow -1/m$, solve

End of Notes